

Task 1: Problem Definition and Approval

- Define a meaningful real-world software quality problem

Embedded computer vision (CV) systems, like those used currently in autonomous robots, drones, and smart cameras, combine resource-strained hardware (microcontrollers, single-board computers) with computationally intensive vision models (such as YOLOv8, or MediaPipe) to make real-time decisions. In lots of development workflows, quality assurance for these systems focuses almost entirely on functional requirements: does the model detect and classify objects accurately? Non-functional requirements such as latency, frame-rate stability, throughput under load, and resource usage (CPU/memory), are frequently treated as inferior concerns, tested improperly or not at all. This is an important gap. A detection that is accurate but arrives too late, or a system that degrades under load, is a genuine failure in a real time context, not just a missed opportunity.

- Explain the background, motivation, stakeholders, users, and practical relevance of the selected problem

- Background

Real time embedded CV is common in robotics, autonomous navigation, robots, and drones, where timing and resource constraints are as critical as detection accuracy. A model that performs well during the benchmark can fail in deployment if it can't keep a consistent frame rate on constrained hardware, or if performance degrades unpredictably as the system load grows. This problem is motivated by direct exposure to this context, where the disconnect between "the model works" and "the system meets its real time requirements" is a recurring, practical concern.

- Stakeholders and Users

Developers/Engineers – Build embedded CV pipelines, who need confidence that functionality doesn't come at the cost of missed timing budgets

QA/Test engineers – Need a defined, repeatable way to test NFRs alongside FRs

End users – Affected if the system misses detections due to latency or degrades under bigger loads

Safety/compliance reviewers – In contexts where delayed or missed detections have real consequences

- Practical Relevance

This is a widely common problem

- **Identify the key software quality issues involved in the problem**
- **Explain why the problem is suitable for a Software Quality Assurance Project**
- **Confirm that the project is achievable within one semester**